

Package ‘mpmaggregate’

March 1, 2026

Type Package

Title Aggregate Matrix Population Models

Version 0.2.0

Date 2026-03-01

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Description Aggregates matrix population models (MPMs) in both the lambda (stable growth rate) and R0 (net reproductive rate) frameworks, including standard and elasticity-consistent aggregators. Standard aggregation in the lambda framework maintains consistent lambda and stable stage distribution, while standard aggregation in the R0 framework maintains consistent R0 and cohort stable stage distribution. Elasticity-consistent aggregators maintain these same consistencies with respect to the chosen framework and additionally preserve consistent reproductive values in the lambda framework and cohort reproductive values in the R0 framework. Aggregation can take the form of general-to-general MPM (mpm_aggregate) or Leslie-to-Leslie MPM (leslie_aggregate).

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Encoding UTF-8

Imports expm

Suggests testthat (>= 3.0.0), roxygen2, knitr, rmarkdown, Rcompadre,
kableExtra, rphylopic

Config/testthat/edition 3

VignetteBuilder knitr

RoxygenNote 7.3.3

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mpmaggregate-package *mpmaggregate: Aggregation of matrix population models*

Description

mpmaggregate aggregates matrix population models (MPMs) under both the lambda (stable growth rate) and R0 (net reproductive rate) frameworks, using either standard or elasticity-consistent aggregation.

Details

The core aggregation approach uses effective left and right operators, P_{eff} and Q_{eff} , to map an original matrix M (dimension n) to an aggregated matrix M_{agg} (dimension m) according to

$$M_{\text{agg}} = P_{\text{eff}} M Q_{\text{eff}}.$$

The operators P_{eff} and Q_{eff} are constructed from the partition matrix P , with definitions depending on the chosen framework and criterion:

- **framework = "lambda"**: aggregation targets the dominant eigenvalue λ and associated eigenvectors.
- **framework = "R0"**: aggregation targets the Perron eigenvalue R_0 of a generation-to-generation matrix derived from reproductive and survival components.
- **criterion = "standard"**: preserves the framework-specific spectral radius (lambda or R0) and stable stage (or cohort stable) distribution.
- **criterion = "elasticity"**: additionally preserves reproductive values (or cohort reproductive values) in the chosen framework.

Aggregation can be performed in two primary settings:

- General-to-general matrix population model aggregation via [mpm_aggregate](#). Groupings are supplied by the user as a list of stage indices (see [mpm_partition](#)).
- Leslie-to-Leslie matrix population model aggregation via [leslie_aggregate](#), which aggregates age classes into cohort blocks and may apply a disaggregation step when required.

The package assumes inputs are nonnegative matrices and (when required for computing Perron eigenvectors used in aggregation) that the population projection matrix is irreducible.

Main functions

- `mpm_aggregate`: general-to-general MPM aggregation using user-defined groups.
- `leslie_aggregate`: Leslie-to-Leslie MPM aggregation into m cohort groups.
- `generation_time`: return the generation time in either the lambda framework or R_0 (cohort-based) framework.
- `mpm_elasticity`: Compute elasticities of either the dominant eigenvalue λ (lambda framework) or the net reproductive rate R_0 (R_0 framework) with respect to the entries of the projection matrix `matA`.

Utility functions

- `mpm_partition`: convert a user-supplied grouping list into a partitioning matrix.
- `spectral_radius`: compute the spectral radius (dominant eigenvalue modulus) of a square matrix.
- `dominant_eigen`: return the dominant eigenvalue (lambda) and associated stable stage distribution and reproductive values for a general population projection matrix.
- `leslie_dominant_eigen`: return the dominant eigenvalue (lambda) and associated stable age distribution and reproductive values for a Leslie population projection matrix.
- `is_leslie`: check whether the input matrix is a Leslie population projection matrix.
- `leslie_disaggregate`: Disaggregate a Leslie population projection model so that its dimensionality is compatible with that of the aggregated model.
- `stable_stage`: Return the stable stage distribution (w) associated with the dominant eigenvalue, scaled so that it sums to one.
- `reproductive_values`: Return the reproductive values (v) associated with the dominant eigenvalue, scaled so that $v*w=1$.
- `leslie_stable_age`: Return the stable age distribution (w) associated with the dominant eigenvalue of a Leslie matrix, scaled so that it sums to one.
- `leslie_reproductive_values`: Return the reproductive values (v) associated with the dominant eigenvalue of a Leslie matrix, scaled so that $v*w=1$.

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References

- Bienvendu, F., Akçay, E., Legendre, S. & McCandlish, D.M., 2017. The genealogical decomposition of a matrix population model with applications to the aggregation of stages. *Theoretical Population Biology*, 115, 69-80. <https://doi.org/10.1016/j.tpb.2017.04.002>
- Bienvendu, F. & Legendre, S., 2015. A new approach to the generation time in matrix population models. *The American Naturalist*, 185(6), 834-843. <https://doi.org/10.1086/681104>
- Caswell, H. 2001. *Matrix population models: construction, analysis and interpretation* (2nd ed.). Sinauer.

Hinrichsen, R. A. 2023. Aggregation of Leslie matrix models with application to ten diverse animal species. *Population Ecology*, 65(3), 146-166. <https://doi.org/10.1002/1438-390X.12149>

Hinrichsen, R. A., Yokomizo, H., & Salguero-Gómez, R. 2026. From theory to application: Elasticity-consistent aggregation of Leslie matrix population models for comparative demography. *bioRxiv*, preprint. <https://doi.org/10.64898/2026.02.04.703802>

Salguero-Gómez, R., & Gamelon, M., eds. 2021. *Demographic methods across the tree of life*. Oxford university press.

Salguero-Gómez, R. & Plotkin, J. B. 2010. Matrix dimensions bias demographic inferences: implications for comparative plant demography. *The American Naturalist* 176, 710-722. <https://doi.org/10.1086/657044>

See Also

[mpm_aggregate](#), [leslie_aggregate](#)

dominant_eigen

Dominant eigen-elements of a population projection matrix

Description

Returns the dominant eigenvalue (lambda) and associated right/left eigenvectors (stable stage distribution w and reproductive values v) for a square population projection matrix A.

Usage

```
dominant_eigen(A, tol = 1e-12, ensure_positive = TRUE)
```

Arguments

A	A square numeric matrix.
tol	Numeric tolerance passed to stable_stage and reproductive_values .
ensure_positive	Logical. Passed to stable_stage and reproductive_values . If TRUE, attempts to orient eigenvectors so entries are mostly positive by flipping sign if needed.

Details

The output is normalized using the conventions in [stable_stage](#) and [reproductive_values](#):

- w is scaled so that $\text{sum}(w) = 1$
- v is scaled so that $\text{sum}(v * w) = 1$

Value

A named list with elements:

- lambda: spectral radius of A
- w: stable stage distribution (right eigenvector), $\text{sum}(w)=1$
- v: reproductive values (left eigenvector), $\text{sum}(v*w)=1$

References

Caswell, H. 2001. Matrix population models: construction, analysis and interpretation (2nd ed.). Sinauer.

See Also

[spectral_radius](#), [stable_stage](#), [reproductive_values](#)

Examples

```
A <- matrix(c(
  0, 1, 2,
  0.5, 0, 0,
  0, 0.8, 0.9
), nrow = 3, byrow = TRUE)
```

```
dom <- dominant_eigen(A)
dom$lambda
sum(dom$w)
sum(dom$v * dom$w)
```

```
#reducible example
A<-rbind(c(1,1,0),
        c(1,0,0),
        c(0,1,0))
```

```
dom <- dominant_eigen(A)
dom$lambda
sum(dom$w)
sum(dom$v * dom$w)
```

GCD

Greatest common divisor of a set of integers

Description

Computes the greatest common divisor (GCD) of a numeric vector *x* of positive integers. Used by [LCM](#).

Usage

```
GCD(x)
```

Arguments

x A numeric vector of positive integers.

Value

A positive integer giving the GCD of *x*.

generation_time	<i>Generation time from a matrix population model</i>
-----------------	---

Description

Returns a generation time for either the λ or R_0 framework

Usage

```
generation_time(
  matF,
  matU,
  matC = NULL,
  framework = c("lambda", "R0"),
  tol = 1e-12
)
```

Arguments

matF	A square fecundity/sexual reproduction matrix (numeric matrix). Must not be NULL.
matU	A square survival/growth/transition matrix (numeric matrix). Must not be NULL.
matC	Optional square clonal reproduction matrix (numeric matrix), same dimensions as matF and matU. Default NULL.
framework	Character scalar. Must be either "lambda" or "R0".
tol	Numeric tolerance used for nonnegativity checks and near-zero screening. Default 1e-8.

Details

The full projection matrix `matA` is constructed internally as `matA = matU + matF (+ matC)` and is not accepted as an input because `matA` alone is not sufficient to define the required decomposition.

Eigen-elements (e.g., λ , R_0 , stable stage distribution, and reproductive values) are intended to be obtained via the workhorse function [dominant_eigen](#).

If `matC` is supplied, the reproductive matrix is defined as `matR = matF + matC`. Otherwise, `matR = matF`.

This function performs input validation, constructs `matA` and `matR`, and sets up the framework-specific branch for computing generation time.

Eigen-related quantities (including λ , R_0 , stable stage distribution, and reproductive values) should be computed using [dominant_eigen](#), which is intended to be the workhorse for those outputs.

Value

A list with elements:

- `framework`: the chosen framework ("lambda" or "R0").
- `generation_time`: generation time (numeric scalar).

References

- Bienvendu, F. & Legendre, S. 2015. A new approach to the generation time in matrix population models. *The American Naturalist*, 185(6), 834-843. <https://doi.org/10.1086/681104>
- Ellner, S. P. 2018. Generation time in structured populations. *The American Naturalist*, 192(1), 105-110. <https://doi.org/10.1086/697539>

See Also

[dominant_eigen](#), [spectral_radius](#).

Examples

```
matU <- matrix(c(0.2, 0.0,
                 0.3, 0.4), nrow = 2, byrow = TRUE)
matF <- matrix(c(0.0, 1.2,
                 0.0, 0.0), nrow = 2, byrow = TRUE)
generation_time(matF = matF, matU = matU, framework = "lambda")
generation_time(matF = matF, matU = matU, framework = "R0")
```

is_leslie

Test whether a matrix is a Leslie matrix

Description

Checks whether A has Leslie form: nonzero entries are allowed only in the first row (fertility rates) and on the subdiagonal (survival probabilities). Survival probabilities must be in $(0, 1]$ (within tolerance), fertility rates must be nonnegative, and at least one fertility rate must be positive.

Usage

```
is_leslie(A, tol = sqrt(.Machine$double.eps))
```

Arguments

A	A numeric square matrix. A length-1 positive numeric will be coerced to a 1x1 matrix.
tol	Numeric tolerance used for structural comparisons and bounds checks.

Details

A 1x1 matrix with a positive entry is treated as a valid (degenerate) Leslie matrix.

Value

Logical; TRUE if A appears to be a Leslie matrix, otherwise FALSE.

Examples

```
L <- matrix(c(
  0, 2, 1,
  0.6, 0, 0,
  0, 0.7, 0
), nrow = 3, byrow = TRUE)
is_leslie(L)
```

LCM	<i>Least common multiple of two integers</i>
-----	--

Description

Computes the least common multiple (LCM) of two positive integers. Used by [leslie_disaggregate](#).

Usage

```
LCM(a, b)
```

Arguments

- a A positive integer.
- b A positive integer.

Value

A positive integer giving the LCM of a and b.

leslie_aggregate	<i>Aggregate Leslie-to-Leslie matrix population model</i>
------------------	---

Description

Aggregates a Leslie matrix to a smaller Leslie matrix by combining ages into `ngroups = m` age groups.

Usage

```
leslie_aggregate(
  matA,
  ngroups,
  framework = c("lambda", "R0"),
  criterion = c("standard", "elasticity"),
  tol = 1e-12,
  ...
)
```


Arguments

matA	A square numeric matrix. Must be finite, nonnegative, and satisfy the structural constraints of a Leslie matrix (checked internally).
ngroups	Positive integer giving the number of aggregated age groups (the dimension m of the aggregated model). Must satisfy $\text{ngroups} \leq n$.
framework	Character string; either "lambda" or "R0". Determines whether aggregation is based on the projection matrix A or an R_0 -based reference matrix built from the implied U and reproductive component of the Leslie matrix.
criterion	Character string; either "standard" or "elasticity". For "elasticity", aggregation uses weights based on reproductive values and the stable age distribution.
tol	Numeric tolerance used in positivity/zero-mass checks and balancing calculations for the elasticity-consistent case.
...	Reserved for future use.

Details

This function assumes the input is a valid Leslie matrix and checks the Leslie structure internally. If the original dimensionality n is not an integer multiple of m , a disaggregation step is applied internally so that the expanded dimensionality is divisible by m before aggregation proceeds.

Aggregation can be performed under two frameworks ("lambda" or "R0") and two criteria ("standard" or "elasticity"), which determine how ages are weighted during aggregation.

Irreducibility requirement. The input Leslie matrix `matA` must be nonnegative, square, and irreducible. This function checks irreducibility of `matA` and will fail if it is reducible. (Any internal disaggregation preserves irreducibility.)

The Leslie matrix is internally decomposed into a reproductive component R (the first row) and a survival component U (the subdiagonal and any survival terms implied by the Leslie structure). For `framework = "R0"`, the next-generation matrix $K = R(I - U)^{-1}$ is formed and its Perron eigenvalue is used as R_0 .

Aggregation proceeds on a k -step matrix where $k = n/m$, with n the dimension of the input Leslie matrix and $m = \text{ngroups}$. The function returns an $m \times m$ k -step Leslie matrix (stored as `matAk_agg`) describing transitions among the aggregated age groups.

The returned element effectiveness is computed for both criteria:

- For `criterion = "standard"`, effectiveness is computed using weights based on the reference stable age distribution w .
- For `criterion = "elasticity"`, effectiveness is computed after balancing transformations (with weights based on $w * v$).

Value

A named list with elements:

`framework` The matched framework used ("lambda" or "R0").

`criterion` The matched criterion used ("standard" or "elasticity").

`matAk_agg` Aggregated k -step Leslie matrix after grouping ($m \times m$).

`effectiveness` A numeric effectiveness measure for the aggregation (definition depends on `criterion`).

References

- Hinrichsen, R. A. 2023. Aggregation of Leslie matrix models with application to ten diverse animal species. *Population Ecology*, 65(3), 146-166. <https://doi.org/10.1002/1438-390X.12149>
- Hinrichsen, R. A., Yokomizo, H., & Salguero-Gómez, R. 2026. From theory to application: Elasticity-consistent aggregation of Leslie matrix population models for comparative demography. *bioRxiv*, preprint. <https://doi.org/10.64898/2026.02.04.703802>

Examples

```
# A simple 3x3 Leslie matrix (fertility rates in first row; survival probabilities on subdiagonal)
A <- matrix(c(
  0.0, 1.2, 1.8,
  0.5, 0.0, 0.0,
  0.0, 0.7, 0.0
), nrow = 3, byrow = TRUE)

# Aggregate to 2 age groups
res_std <- leslie_aggregate(
  matA = A,
  ngroups = 2,
  framework = "lambda",
  criterion = "standard"
)
res_std$matAk_agg
res_std$effectiveness

res_el <- leslie_aggregate(
  matA = A,
  ngroups = 2,
  framework = "lambda",
  criterion = "elasticity"
)
res_el$matAk_agg
res_el$effectiveness
```

leslie_disaggregate	<i>Disaggregate a Leslie matrix population model to a compatible dimensionality</i>
---------------------	---

Description

Expands an $n \times n$ Leslie matrix A to an $n_{new} \times n_{new}$ Leslie matrix, where $n_{new} = \text{LCM}(n, m)$.

Usage

```
leslie_disaggregate(A, m)
```

Arguments

- | | |
|-----|---|
| A | A Leslie matrix (checked with is_leslie). A length-1 positive numeric is treated as a 1x1 Leslie matrix. |
| m | Target aggregated dimensionality (positive integer). |

Details

Disaggregation is required when aggregating a Leslie model to dimensionality m and m does not divide n evenly. The expanded matrix introduces a finer age structure that is compatible with both dimensionalities.

Each original age class is subdivided into $m_1 = n_{new}/n$ sub-classes. Fertility rates from original age class i are placed at the end of the corresponding block, deterministic aging within blocks is represented by ones on the subdiagonal, and original survival probabilities are inserted at block boundaries.

Value

A Leslie matrix of dimensionality $n_{new} \times n_{new}$, where $n_{new} = \text{LCM}(n, m)$.

References

Hinrichsen, R. A. 2023. Aggregation of Leslie matrix models with application to ten diverse animal species. *Population Ecology*, 65(3), 146-166. <https://doi.org/10.1002/1438-390X.12149>

Hinrichsen, R. A., Yokomizo, H., & Salguero-Gómez, R. 2026. From theory to application: Elasticity-consistent aggregation of Leslie matrix population models for comparative demography. *bioRxiv*, preprint. <https://doi.org/10.64898/2026.02.04.703802>

Examples

```
L <- matrix(c(
  0, 2, 1,
  0.6, 0, 0,
  0, 0.7, 0
), nrow = 3, byrow = TRUE)
leslie_disaggregate(L, m = 2)
```

leslie_dominant_eigen *Dominant eigen-elements of a Leslie matrix*

Description

Returns the dominant eigenvalue (lambda) and the stable age distribution w and reproductive values v for a Leslie matrix A .

Usage

```
leslie_dominant_eigen(A)
```

Arguments

A A Leslie matrix (checked with `is_leslie`). A length-1 positive numeric will be coerced to a 1x1 matrix.

Details

Normalization: #' The output is normalized using the conventions in [leslie_stable_age](#) and [leslie_reproductive_values](#)

- w is scaled so that $\text{sum}(w) = 1$
- v is scaled so that $\text{sum}(v * w) = 1$

A length-1 positive numeric is treated as a 1x1 Leslie matrix and returns $\lambda = A$, $w = 1$, $v = 1$.

Value

A named list with elements:

- lambda: spectral radius of A
- w: stable age distribution, $\text{sum}(w)=1$
- v: reproductive values, $\text{sum}(v*w)=1$

References

Caswell, H. 2001. Matrix population models: construction, analysis and interpretation (2nd ed.). Sinauer.

Demetrius, L. 1974. Demographic parameters and natural selection. Proceedings of the National Academy of Sciences, 71(12), 4645-4647. <https://doi.org/10.1073/pnas.71.12.4645>

See Also

[spectral_radius](#), [leslie_stable_age](#), [leslie_reproductive_values](#)

Examples

```
leslie_dominant_eigen(1)

L <- matrix(c(
  0, 2, 1,
  0.6, 0, 0,
  0, 0.7, 0
), nrow = 3, byrow = TRUE)
dom <- leslie_dominant_eigen(L)
dom$lambda
sum(dom$w)
sum(dom$v * dom$w)
```

leslie_reproductive_values

Reproductive values for a Leslie matrix

Description

Computes the reproductive value vector v for a Leslie matrix A, scaled so that $\text{sum}(v * w) = 1$, where w is the stable age distribution returned by [leslie_stable_age](#).

Usage

```
leslie_reproductive_values(A)
```

Arguments

A A Leslie matrix (checked with `is_leslie`). A length-1 positive numeric will be coerced to a 1x1 matrix.

Details

This implementation follows the Demetrius (1974) shortcut expressed in terms of the stable age distribution and fertility rates.

A length-1 positive numeric is treated as a 1x1 Leslie matrix and returns 1.

Value

A numeric vector of length `nrow(A)` giving reproductive values, normalized so that $\text{sum}(v * w) = 1$.

References

Caswell, H. 2001. Matrix population models: construction, analysis and interpretation (2nd ed.). Sinauer.

Demetrius, L. 1974. Demographic parameters and natural selection. Proceedings of the National Academy of Sciences, 71(12), 4645-4647. <https://doi.org/10.1073/pnas.71.12.4645>

Examples

```
leslie_reproductive_values(1)

L <- matrix(c(
  0, 2, 1,
  0.6, 0, 0,
  0, 0.7, 0
), nrow = 3, byrow = TRUE)
v <- leslie_reproductive_values(L)
w <- leslie_stable_age(L)
sum(v * w) # should be 1
```

leslie_stable_age	<i>Stable age distribution for a Leslie matrix</i>
-------------------	--

Description

Computes the stable age distribution for a Leslie matrix A using the standard recursion based on subdiagonal survival probabilities and the dominant eigenvalue (spectral radius). The output is scaled so that $\text{sum}(w) = 1$.

Usage

```
leslie_stable_age(A)
```

Arguments

A A Leslie matrix (checked with `is_leslie`). A length-1 positive numeric will be coerced to a 1x1 matrix.

Details

A length-1 positive numeric is treated as a 1x1 Leslie matrix and returns 1.

Value

A numeric vector of length `nrow(A)` giving the stable age distribution, scaled so that `sum(w) = 1`.

References

Caswell, H. 2001. Matrix population models: construction, analysis and interpretation (2nd ed.). Sinauer.

Demetrius, L. 1974. Demographic parameters and natural selection. Proceedings of the National Academy of Sciences, 71(12), 4645-4647. <https://doi.org/10.1073/pnas.71.12.4645>

Examples

```
leslie_stable_age(1)

L <- matrix(c(
  0, 2, 1,
  0.6, 0, 0,
  0, 0.7, 0
), nrow = 3, byrow = TRUE)
leslie_stable_age(L)
```

mpm_aggregate

Aggregate a general-to-general matrix population model

Description

Aggregates one or more matrix population model (MPM) components from an $n \times n$ model to an $m \times m$ model by combining stages into user-defined groups.

Usage

```
mpm_aggregate(
  matA = NULL,
  matU = NULL,
  matF = NULL,
  matC = NULL,
  groups,
  framework = c("lambda", "R0"),
  criterion = c("standard", "elasticity"),
  tol = 1e-12,
  ...
)
```

Arguments

matA	Optional projection matrix A (square, finite, nonnegative). If NULL, A is constructed as $U + (F + C)$.
matU	Optional survival-transition matrix U (square, finite, nonnegative). Required when framework = "R0" and/or when matA is NULL.
matF	Optional fecundity matrix F (square, finite, nonnegative). Interpreted as part of reproduction $R = F + C$.
matC	Optional clonal reproduction matrix C (square, finite, nonnegative). Treated as reproduction and combined with matF as $R = F + C$.
groups	A non-empty list of integer vectors specifying aggregation groups. Each stage in 1:n must appear exactly once across the list.
framework	Character string; either "lambda" or "R0". Determines whether aggregation is based on the projection matrix A or an R_0 -based reference matrix.
criterion	Character string; either "standard" or "elasticity". For "elasticity", aggregation uses weights based on reproductive values and the stable stage distribution.
tol	Numeric tolerance used in positivity/zero-mass checks and balancing calculations for the elasticity-consistent case.
...	Reserved for future use.

Details

Groupings are supplied by groups, a list of integer vectors specifying how original stages (1,...,n) are combined; see [mpm_partition](#). Each original stage must appear exactly once.

Aggregation can be performed under two frameworks ("lambda" or "R0") and two criteria ("standard" or "elasticity"), which determine how stages are weighted during aggregation.

Clonal reproduction matC is treated as reproductive output. When both matF and matC are supplied, the effective reproductive matrix $R = F + C$ is used internally (e.g., in the "R0" framework) but is not returned. The aggregated effective reproduction can be recovered as matF_agg + matC_agg.

Irreducibility requirement. The effective projection matrix A must be nonnegative, square, and irreducible. This function enforces irreducibility of A as either matA (if supplied) or $A = U + (F + C)$ otherwise, and will fail if reducible.

When framework = "R0", this function requires matU and at least one of matF or matC in order to form the effective reproductive matrix $R = F + C$ and the generation-to-generation matrix $K = R(I - U)^{-1}$. The matrix R is used internally for computing reference quantities in the "R0" framework but is not returned. The aggregated effective reproduction can be obtained as matF_agg + matC_agg when both components are supplied.

All returned matrices are aggregated using the same weighting rules implied by the selected framework and criterion.

The returned element effectiveness is computed for both criteria:

- For criterion = "standard", effectiveness is computed using weights based on the reference stable stage distribution w .
- For criterion = "elasticity", effectiveness is computed after balancing transformations (with weights based on $w * v$).

Value

A named list with elements:

`framework` The matched framework used ("lambda" or "R0").

`criterion` The matched criterion used ("standard" or "elasticity").

`matA_agg` Aggregated projection matrix A_{agg} .

`matU_agg` Aggregated survival-transition matrix U_{agg} , or NULL if `matU` was not supplied.

`matF_agg` Aggregated fecundity matrix F_{agg} , or NULL if `matF` was not supplied.

`matC_agg` Aggregated clonal reproduction matrix C_{agg} , or NULL if `matC` was not supplied.

`effectiveness` Numeric effectiveness measure for the aggregation (definition depends on `criterion`).

References

Bienvenu, F., Akcay, E., Legendre, S. and McCandlish, D.M. 2017. The genealogical decomposition of a matrix population model with applications to the aggregation of stages. *Theoretical Population Biology*, 115, 69-80. <https://doi.org/10.1016/j.tpb.2017.04.002>

Hooley, D. E. 2000. Collapsed matrices with (almost) the same eigenstuff. *The College Mathematics Journal*, 31(4), 297-299. <https://doi.org/10.1080/07468342.2000.11974162>

Salguero-Gomez, R. & Plotkin, J. B. 2010. Matrix dimensions bias demographic inferences: implications for comparative plant demography. *The American Naturalist* 176, 710-722. <https://doi.org/10.1086/657044>

Examples

```
# Example aggregation of a 3x3 projection matrix to 2x2 using groups:
# group 1 = stage 1; group 2 = stages 2 and 3.
A <- matrix(c(
  0.0, 1.0, 2.0,
  0.5, 0.0, 0.0,
  0.0, 0.8, 0.9
), nrow = 3, byrow = TRUE)

res_std <- mpm_aggregate(
  matA = A,
  groups = list(c(1), c(2, 3)),
  framework = "lambda",
  criterion = "standard"
)
res_std$matA_agg
res_std$effectiveness

res_el <- mpm_aggregate(
  matA = A,
  groups = list(c(1), c(2, 3)),
  framework = "lambda",
  criterion = "elasticity"
)
res_el$matA_agg
res_el$effectiveness
```

mpm_elasticity	<i>Elasticity of λ or R_0 with respect to entries of matA</i>
----------------	---

Description

Compute elasticities of either the dominant eigenvalue λ (lambda framework) or the net reproductive rate R_0 (R_0 framework) with respect to the entries of the projection matrix matA.

Usage

```
mpm_elasticity(
  matA = NULL,
  matF = NULL,
  matU = NULL,
  matC = NULL,
  framework = c("lambda", "R0"),
  normalize = TRUE,
  tol = 1e-12
)
```

Arguments

matA	A square projection matrix (numeric matrix). Required when framework = "lambda". Must be NULL when framework = "R0".
matF	A square fecundity/sexual reproduction matrix (numeric matrix). Required when framework = "R0". Must be NULL when framework = "lambda".
matU	A square survival/growth/transition matrix (numeric matrix). Required when framework = "R0". Must be NULL when framework = "lambda".
matC	Optional square clonal reproduction matrix (numeric matrix). Only allowed when framework = "R0".
framework	Character scalar. Must be either "lambda" or "R0".
normalize	Logical. Must be either TRUE or FALSE. This is relevant only when framework = "R0". When true, the elasticity of "R0" with respect to the entries of "matA" are scaled to sum to 1.
tol	Numeric tolerance used for nonnegativity checks. Default 1e-12.

Details

The function enforces a strict separation of inputs by framework:

- framework = "lambda": matA must be provided and matF, matU, and matC must be NULL.
- framework = "R0": matF and matU must be provided, matC is optional, and matA must be NULL. The projection matrix is constructed internally as matA = matR + matU, where matR = matF (+ matC).

Irreducibility of the resulting matA is enforced using the internal helper `.check_irreducible_hj` (defined elsewhere in the package).

Eigen-elements (e.g., dominant eigenvalue, left and right eigenvectors) should be obtained via the workhorse function [dominant_eigen](#).

Lambda framework. Elasticity is the elasticity of λ with respect to the entries of `matA`.

R0 framework. Elasticity is the elasticity of R_0 with respect to the entries of the internally constructed `matA`.

Value

A list with elements:

- `framework`: the chosen framework.
- `elasticity`: a matrix of elasticities with the same dimensionality as `matA`.

References

Caswell, H. 2001. Matrix population models: construction, analysis and interpretation (2nd ed.). Sinauer.

See Also

[dominant_eigen](#), [spectral_radius](#)

Examples

```
## Lambda framework: matA provided directly
matA <- matrix(
  c(0.2, 1.2,
    0.3, 0.4),
  nrow = 2, byrow = TRUE
)
out_lambda <- mpm_elasticity(matA = matA, framework = "lambda")
str(out_lambda)

## R0 framework: matA constructed from matF and matU
matU <- matrix(
  c(0.2, 0.0,
    0.3, 0.4),
  nrow = 2, byrow = TRUE
)
matF <- matrix(
  c(0.0, 1.2,
    0.0, 0.0),
  nrow = 2, byrow = TRUE
)
out_R0 <- mpm_elasticity(matF = matF, matU = matU, framework = "R0")
str(out_R0)
```

mpm_partition

Create a partitioning matrix for MPM aggregation

Description

Builds a 0/1 partitioning matrix P that maps an n -stage model to an m -stage aggregated model by summing specified original stages within each aggregated stage.

Usage

```
mpm_partition(groups, n = NULL)
```

Arguments

groups A list of integer vectors giving stage indices for each aggregated group.
n Optional integer. Number of original stages. If NULL, inferred as `max(unlist(groups))`.

Details

The grouping is provided as a list of integer vectors. For example, `groups = list(c(1), c(2:3))` defines $m = 2$ aggregated stages: the first contains original stage 1, and the second contains original stages 2 and 3.

Each original stage index `1:n` must appear **once and only once** across groups.

Value

A numeric matrix P of dimensionality $m \times n$ with entries 0 or 1.

Examples

```
g <- list(c(1), c(2:3))
P <- mpm_partition(g, n = 3)
P
```

reproductive_values	<i>Compute reproductive values</i>
---------------------	------------------------------------

Description

Returns the reproductive value vector v for a square nonnegative matrix A , defined as the left eigenvector associated with the dominant eigenvalue (spectral radius). The vector is normalized in the usual demographic way so that $\text{sum}(v * w) = 1$, where w is the stable stage distribution returned by [stable_stage](#).

Usage

```
reproductive_values(A, tol = 1e-12, ensure_positive = TRUE)
```

Arguments

A A square numeric nonnegative matrix.
tol Numeric tolerance used for normalization/positivity checks.
ensure_positive Logical. If TRUE, attempts to orient the eigenvector so its entries are mostly positive by flipping its sign if necessary.

Details

For matrix population models with nonnegative irreducible matrix, reproductive values are strictly positive (up to scaling).

Value

A numeric vector of length `nrow(A)` giving reproductive values, scaled so that $\text{sum}(v * w) = 1$.

References

Caswell, H. 2001. Matrix population models: construction, analysis and interpretation (2nd ed.). Sinauer.

Examples

```
A <- matrix(c(
  0, 1,
  0.5, 0
), nrow = 2, byrow = TRUE)

w <- stable_stage(A)
v <- reproductive_values(A)
v
sum(v * w) # should be 1

A<-rbind(c(1,1,0),
          c(1,0,0),
          c(0,1,0))

w <- stable_stage(A)
w
sum(w)

w <- stable_stage(A)
v <- reproductive_values(A)
v
sum(v * w) # should be 1
```

spectral_radius

Compute the spectral radius of a matrix

Description

Returns the spectral radius of a square matrix, defined as the maximum modulus of its eigenvalues:

$$\rho(A) = \max |\lambda_i|$$

Usage

```
spectral_radius(A)
```

Arguments

A A square numeric matrix.

Details

This function is primarily intended for use with matrix population models, where the spectral radius corresponds to the asymptotic growth rate (i.e., λ) when the governing matrix is square and nonnegative. It is also used to calculate net reproductive rate R_0 from a generation-to-generation matrix.

Value

A single numeric value giving the spectral radius of A.

References

Horn, R.A. and Johnson, C.R. 2013. Matrix analysis. Cambridge university press.

Examples

```
A <- matrix(c(
  0, 1,
  0.5, 0
), nrow = 2, byrow = TRUE)

spectral_radius(A)
```

stable_stage	<i>Compute the stable stage distribution</i>
--------------	--

Description

Returns the stable stage distribution for a square nonnegative matrix A, defined as the right eigenvector associated with the dominant eigenvalue. The returned vector w is normalized so that $\text{sum}(w) = 1$.

Usage

```
stable_stage(A, tol = 1e-12, ensure_positive = TRUE)
```

Arguments

A	A square numeric matrix.
tol	Numeric tolerance used for positivity/normalization checks.
ensure_positive	Logical. If TRUE, attempts to orient the eigenvector so its entries are mostly positive by flipping its sign if necessary.

Details

For matrix population models with nonnegative irreducible matrix, stable stage densities are strictly positive (up to scaling).

Value

A numeric vector of length `nrow(A)` giving the stable stage distribution, normalized to sum to 1.

References

Caswell, H. 2001. Matrix population models: construction, analysis and interpretation (2nd ed.). Sinauer.

Examples

```
#irreducible example
A <- matrix(c(
  0, 1,
  0.5, 0
), nrow = 2, byrow = TRUE)

w <- stable_stage(A)
w
sum(w)
#reducible example
A<-rbind(c(1,1,0),
        c(1,0,0),
        c(0,1,0))

w <- stable_stage(A)
w
sum(w)
```

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